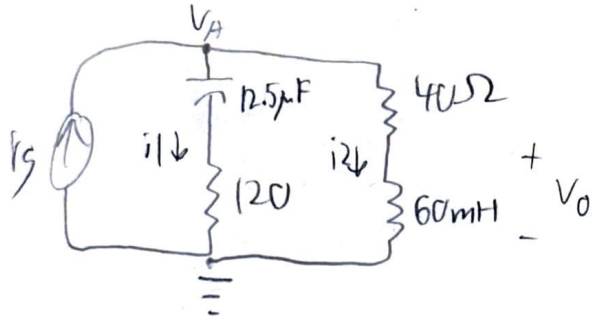
9.32

Rylan Tromley



12.5μF to $Z = -j40$

60mH to $Z = j120$

$i_g = .5\cos(2000t) = .5\angle 0 = .5$

$-i_g + i_1 + i_2 = 0$

$i_1 = \frac{V_a}{120 - j40}$

$i_2 = \frac{V_a}{40 + j120}$

$-\frac{1}{2} + \frac{V_a}{120 - j40} + \frac{V_a}{40 + j120} = 0$

$V_a\left(\frac{1}{120 - j40} + \frac{1}{40 + j120}\right) = \frac{1}{2}$

$V_a = 40 + 20j$

$\frac{40 + 20j}{40 + j120} = i_2$

$i_2 = .25 - .25j$

$j120(.25 - .25j) = 30 + 30j$

$30^2 + 30^2 = m^2$

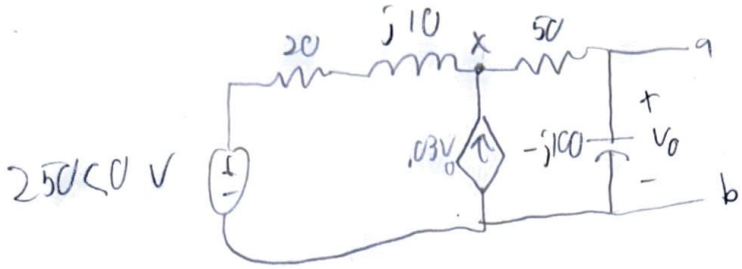
$42.43 = m$

$\tan^{-1}\left(\frac{30}{30}\right) = 45$

$42.43\angle 45$

$\boxed{V_o = 42.43\cos(2000t + 45)\ V}$

9.48



$$\frac{V_x - 250}{20+j10} - .03V_o + \frac{V_x}{50-j100} = 0$$

$$V_x\left(\frac{-j100}{50-j100}\right) = V_o$$

$$\frac{V_x}{20+j10} - \frac{250}{20+j10} + \frac{3jV_x}{50-j100} + \frac{V_x}{50-j100} = 0$$

$$V_x\left(\frac{1}{20+j10} + \frac{3j}{50-j100} + \frac{1}{50-j100}\right) = \frac{250}{20+j10}$$

$$V_x = 500-250j$$

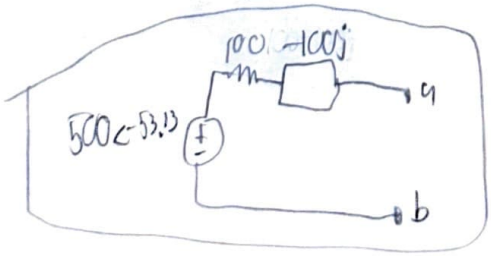
$$V_o = 300-400j$$

$$V_o = V_{th}$$

$$V_{th} = 300-400j$$

$$\frac{250}{20+50+j10} = I_{sc} = 3.5-.5j$$

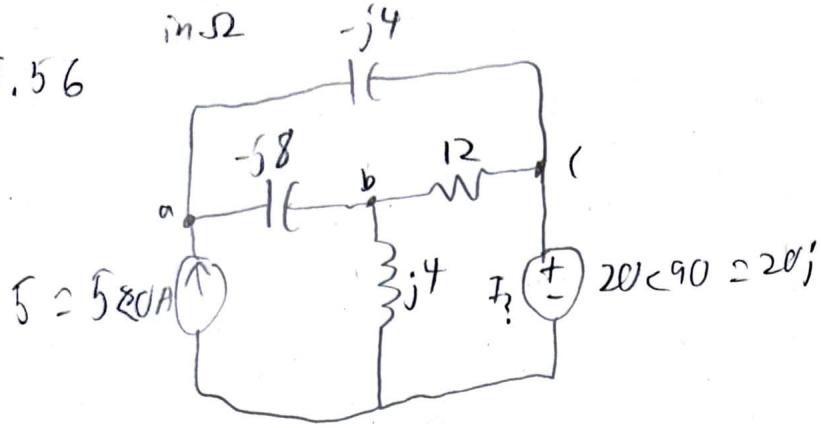
$$\frac{300-400j}{3.5-.5j} = 100-100j \; R_{2th}$$



$$\tan^{-1}\left(\frac{-400}{300}\right) = -53.13^\circ$$

$500<-53.13$

9.56



a

$$-5 + \frac{V_a - V_b}{-j8} + \frac{V_a - V_c}{-j4} = 0$$

b

$$\frac{V_b - V_a}{-j8} + \frac{V_b - V_c}{12} + \frac{V_b}{j4} = 0 \qquad V_c = 20j$$

c

$$\frac{V_c - V_b}{12} + \frac{V_c - V_a}{-j4} + I_? = 0$$

← don't need for this problem luckily

$$V_a = -\frac{8}{3} + \frac{4j}{3}\ V$$

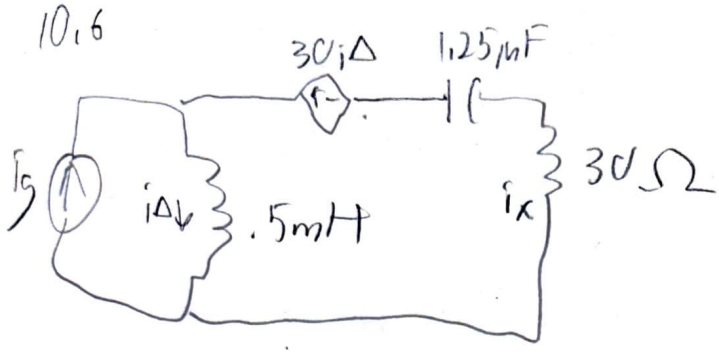
$$V_b = -8 + 4j\ V$$

$$V_g = V_a - V_c$$

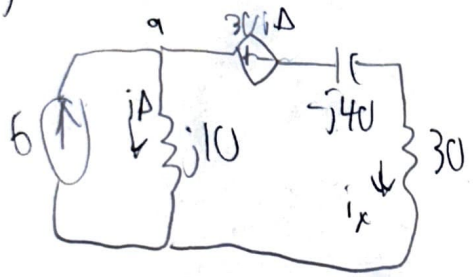
$$-\frac{8}{3} + \frac{4j}{3} - 20j = \boxed{-\frac{8}{3} - \frac{56}{3}j\ V = V_g}$$

$$\boxed{V_g = 18.86\angle -98.13}$$

10.6



$i_g = 6\cos 200000t = 6A$



$$\frac{V_a}{j10} = i_\Delta$$

$$\frac{30(V_a)}{j10}$$

$$-6 + \frac{V_a}{j10} + \frac{V_a}{30-j40} - \frac{30\left(\frac{V_a}{j10}\right)}{30-j40} = 0$$

$$\frac{V_a(30-j40)}{j10} + V_a - 30\left(\frac{V_a}{j10}\right) = 180-240j$$

$$V_a(30-j40) + V_a j10 - 30V_a = 1800j + 2400$$

$$V_a = \frac{1800j + 2400}{30-j40+j10}$$

$$V_a = 60-80j$$

$$V_a = 100\angle -53.13$$

$$\frac{60-80j}{j10} = i_\Delta = 8+6j$$

$$i_g = i_\Delta + i_x$$

$$i_g - i_\Delta = i_x$$

$$6 - 8 + 6j = -2+6j$$

$$\frac{1}{2}|-2+6j|^2\, 30$$

$$\frac{1}{2}(40)30 = 600$$

$\boxed{\text{600 watts absorbed}}$